How to debug a Python CLI package in Visual Studio Code?

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I'm developing a Python project that runs in the CLI using the following command:

```
my-python-package --config config.yaml
```

Recently, some dependencies were updated, leading to deprecated or eliminated functions that I need to debug.

I’m using Visual Studio Code (VS Code) and have attempted to create a `launch.json` configuration, but it only works when running a specific Python file directly. My project does not have a `main.py`, and executing the entry point does not produce any results.

My goal is to run the CLI command and debug it visually in VS Code, similar to how one would debug a regular Python file.

I have tried various configurations in `launch.json`, as suggested in How to debug a Python module in Visual Studio Code's launch.json, but none have worked since my project is not structured as a traditional module. I also asked ChatGPT for help, but it continues to suggest executing the entry point, which does not work either.

Project tree

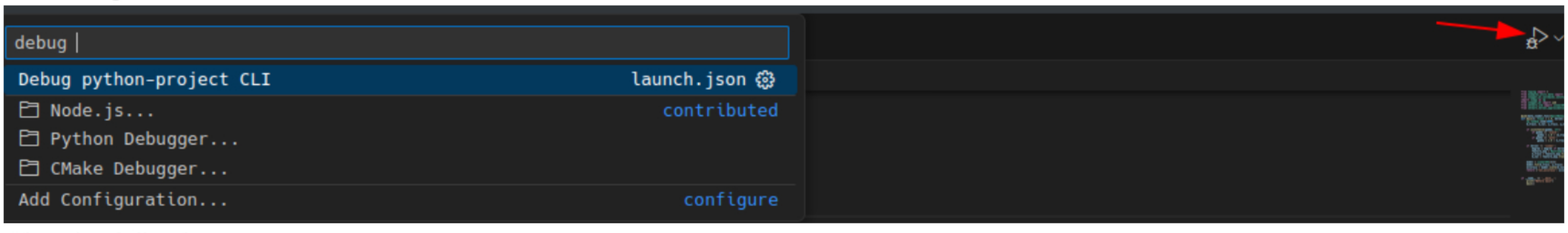
```
my-project-folder
├── __init__.py
├── callbacks
│   ├── __init__.py
│   ├── core.py
│   └── notifiers.py
├── config
│   ├── __init__.py
│   └── models.py
├── exceptions.py
├── experiments.py
├── mlflow.py
├── optuna
│   ├── __init__.py
│   └── types.py
├── scripts
│   ├── __init__.py
│   └── run_experiment.py <- This is the entry point in setup.py
└── utils.py
example.py
```

Output

Trying with the following launch.json

```
{
   "version": "0.2.0",
   "configurations": [
       {
           "name": "Debug python project CLI",
           "type": "debugpy",
           "request": "launch",
           "module": "project_name_folder",
           "console": "integratedTerminal",
           "cwd": "${workspaceFolder}",
           "env": {},
           "envFile": "${workspaceFolder}/.env",
           "justMyCode": true
       }
   ]
}
```

Executing code



Give the following error:

```
> cd .../projectName ; /usr/bin/env ../projectName/.venv/bin/python .../.vscode/extensions/ms-python.debugpy

.../ProjectName/.venv/bin/python: No module named ProjectName.__main__; 'project-name' is a package and cann
```

Development process:

1. I installed the package locally for the first time using the following commands:

   ```
   python -m venv .venv && source .venv/bin/activate && pip install -e .
   ```

2. Then I modify the code in VSC and run the CLI trying to fix the errors.

Current `launch.json` configuration:

Here's my current `launch.json`, which works for a specific example file:

```
{
   "version": "0.2.0",
   "configurations": [
       {
        "name": "Python Debugger: train_svm.py file",
        "type": "debugpy",
        "request": "launch",
        "program": "example.py",
        "console": "integratedTerminal",
        "args": [
           "examples/experiments/config.yaml"
        ],
        "justMyCode": true
     }
   ]
}
```

However, when I try to run the CLI command, I encounter different errors.

EDIT

I posted my solution in How can I debug Python console_script command line apps with the VSCode debugger?

python visual-studio-code command-line-interface

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edited Oct 8 at 15:48

asked Oct 6 at 17:44
Yerasito
3 •6

1 You may check this out: stackoverflow.com/questions/64556874/… . – Amadeusz Liu Oct 7 at 7:39

Adding the idiomatic if **name** == '**main**': app() to the entry point file worked. However, I still get the same errors as when I run the Python file directly. Meanwhile, using the CLI command results in different errors. Thanks for your comment, that was useful anyways. – Yerasito Oct 7 at 15:31

@AmadeuszLiu Should I mark this question as duplicated and post my solution in your suggested stackoverflow post? – Yerasito Oct 7 at 16:22

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